

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B083-Networkwalks
Date	18 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers two practical modules completed during Week 2 of my Cybersecurity internship at Networkwalks. The first module (W2-PM1) covers footprinting the networkwalks.com domain using multiple Kali Linux tools. The Second module (W2-PM5) covers scanning my own local network with Zenmap. Together, these three activities show how an attacker (or a defender testing their own exposure) moves from gathering public

information, to searching for already-indexed exposed assets, to actively mapping live hosts on a network.

All footprinting commands were run in Kali Linux inside a VirtualBox virtual machine. The network scan was performed on a Windows PC with Zenmap installed. Every step below includes the exact command or dork used, the result observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose for Activities Performed.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and scanning Activities
WHOIS	Find domain registration details (owner, dates, name servers)
WhatWeb	Fingerprint web technologies (server, CMS, plugins, IP)
Nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers of the website
Wafw00f	Detect whether a Web Application Firewall protects the site
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses
Windows CMD	Local IP address and MAC address identification

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f. Each tool was used to collect a different type of information about the target.

First, I used WHOIS to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration

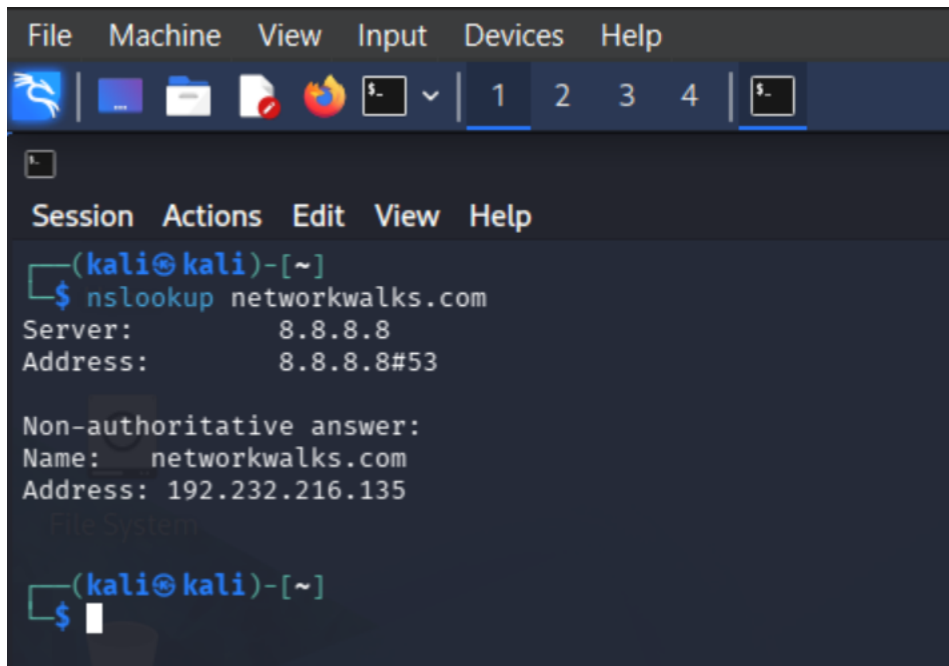
and hosting infrastructure.

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)~  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-09-17T10:21:56Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration
```

I then used WhatWeb to identify technologies used by the website. The results identified WordPress 7.0.4 and WP Download Manager 3.3.58, along with other information exposed by the website.

```
File Machine View Input Devices Help  
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)~  
$ whatweb networkwalks.com  
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[[_wpdm_client], Country[UNITED STATES][en], HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]  
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[_wpdm_client], Country[UNITED STATES][en], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[468ANR-IPIB&pid=50511&prt=76&scv=76&scvVar2=12,application/json,application/ld+json,module.speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]  
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][en], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[468ANR-IPIB&pid=50511&prt=76&scv=76&scvVar2=12,application/json,application/ld+json,module.speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```

Using Nslookup, I resolved the domain name to its IP address. The provided result identified 192.232.216.135.



```
File Machine View Input Devices Help
[Icons] | 1 2 3 4 | [Terminal Icon]

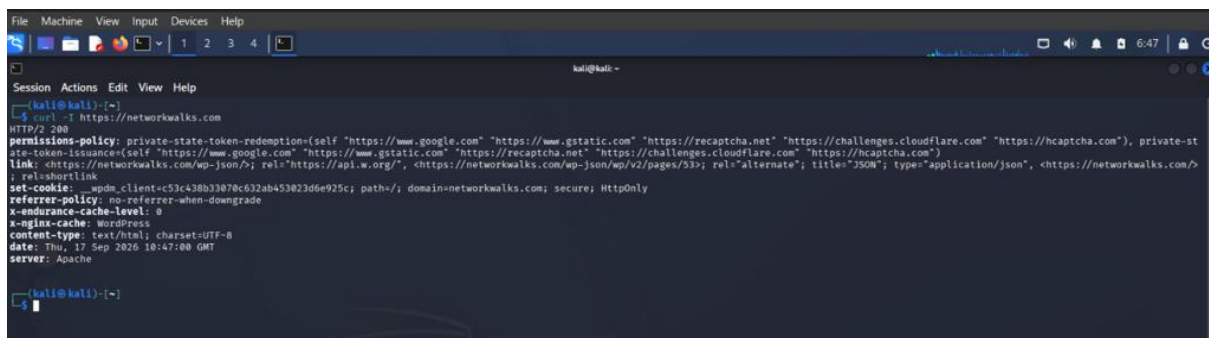
Session Actions Edit View Help

(kali@kali)-[~]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135

(kali@kali)-[~]
$
```

I used Curl with the -I option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint /wp-json/.

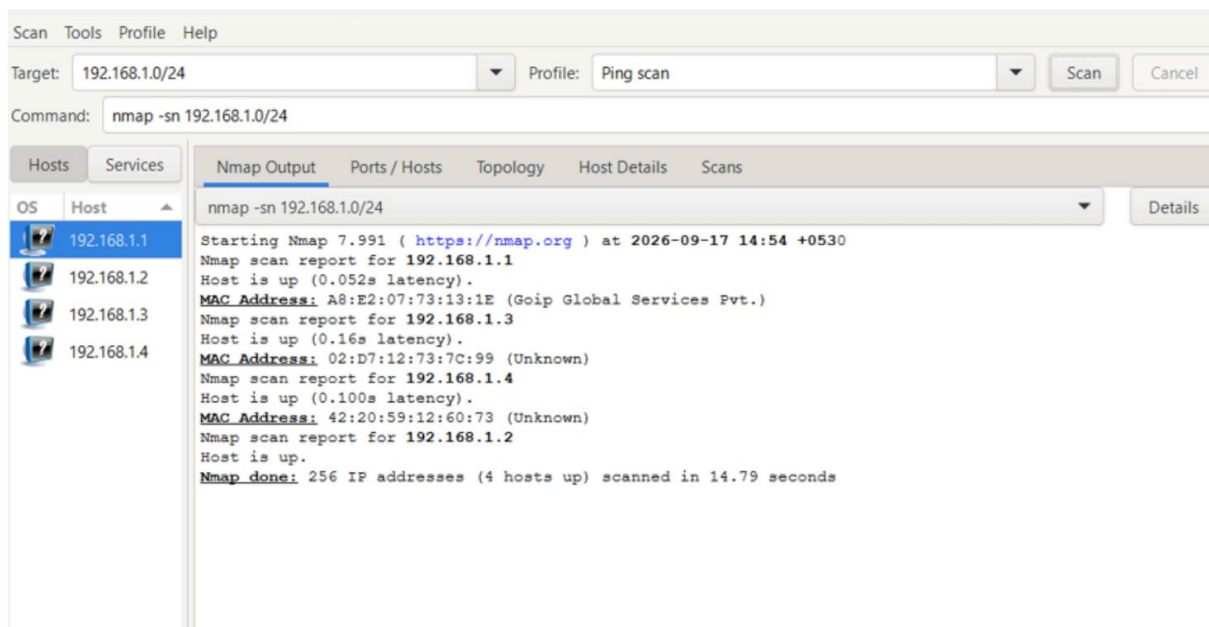
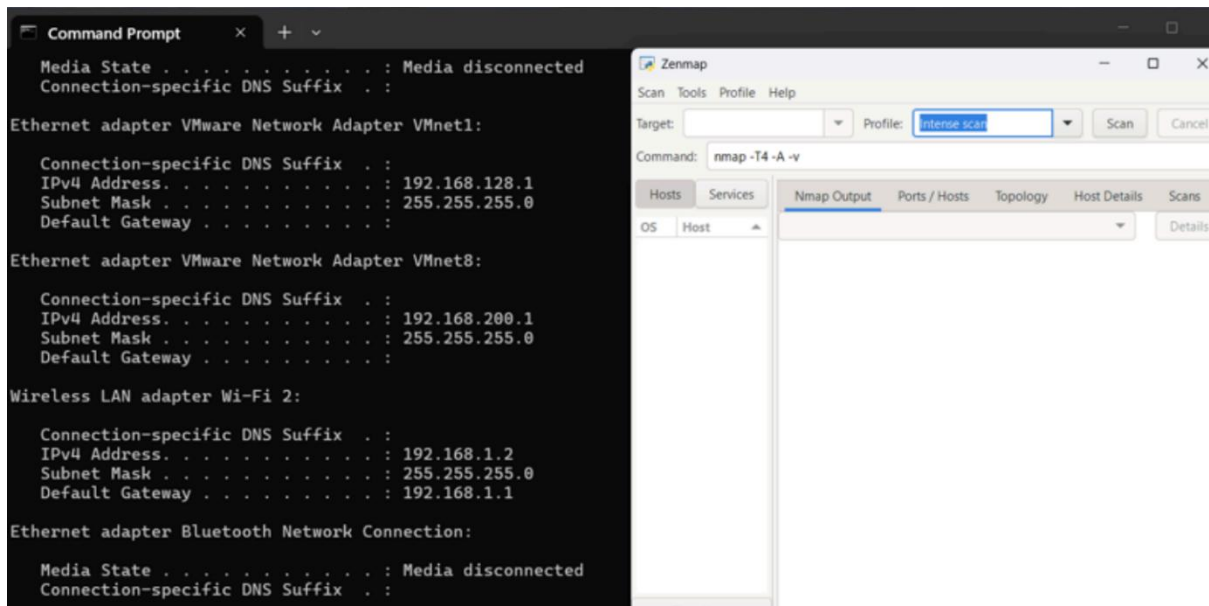


```
File Machine View Input Devices Help
[Icons] | 1 2 3 4 | [Terminal Icon]

kali@kali:~$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: wpdm_client=c53c420b33070c632ab453023d6e925c; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Thu, 17 Sep 2026 10:47:00 GMT
server: Apache

(kali@kali)-[~]
$
```

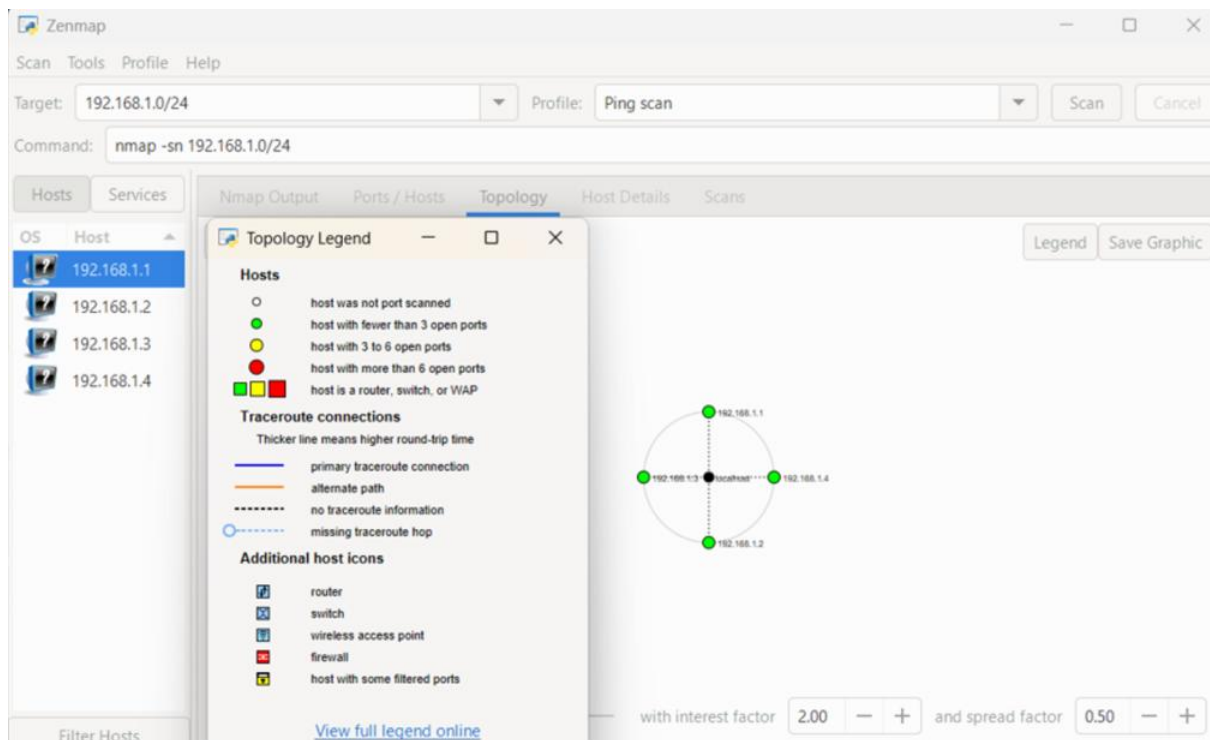
Next, I used Wafw00f to determine whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs).



Task Results

- Number of live hosts found: 4 (including my own PC)
- Live host IP addresses: 192.168.1.1, 192.168.1.2, 192.168.1.3, 192.168.1.4
- MAC addresses identified: A8:E2:07:73:13:1E (192.168.1.1 - Goip Global Services Pvt.), 02:D7:12:73:7C:99 (192.168.1.3 - unknown vendor), 42:20:59:12:60:73 (192.168.1.4 - unknown vendor)
- 192.168.1.2 is my own PC, so no external MAC address was reported for it by the scan

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.



Note: The actual subnet, number of hosts and addresses should be replaced with the results from my own network when submitting the report.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1 and WP Download Manager 3.3.58	Attackers may use exposed technology/version information to identify software requiring further security review	Medium
2	Server IP address identifiable	Nslookup and WhatWeb resolved the domain to 192.232.216.135	Reveals the network of the web service	Low
3	HTTP technical information exposed	curl -I returned response headers and exposed the /wp-json/ REST API endpoint	May assist in technology footprinting and further enumeration	Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the site's security architecture	Low
5	DNS infrastructure information exposed	DNSRecon identified NS/MX/SPF/TXT/SRV records and BIND software versions on both name servers	DNS and software-version information can help build a broader infrastructure and attack profile	Medium
6	Multiple live hosts visible on local network	Zenmap identified 4 live hosts with resolvable MAC addresses on my home network	Unknown or unauthorised devices may potentially be present on a network	Medium

Risk level key: ● Critical ● Medium ●

The risks above are observations from footprinting, GHDB searching and host discovery exercises, not confirmed vulnerabilities on any system I do not own or control. No exploitation or vulnerability validation was performed as part of these modules. The presence of information such as a software version, IP address, DNS record, or an internet-indexed device does not by itself prove a system is exploitable; further authorised security testing would be required to confirm any actual vulnerability.

6. Recommendations

- Review publicly exposed technology information: organisations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
- Keep software updated: CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
- Review HTTP headers: response headers should be checked to ensure unnecessary technical information is not being exposed.
- Review DNS records regularly: DNS records should be checked periodically to ensure only required information and services are publicly exposed, and default software banners (e.g. BIND version) should be suppressed where possible.
- Properly configure and monitor the WAF: keep ModSecurity enabled and tuned, since it already blocks naive attacks.
- Never expose administrative or camera interfaces without authentication: default-credential or no-auth camera panels should never be reachable directly from the internet; place them behind a VPN or authenticated gateway.
- Disable directory listing on web servers: open, browsable directories should be turned off so files are only reachable through an intended, access-controlled path.
- Perform regular internal network discovery: periodically scan your own network to identify active devices and investigate anything unexpected.
- Maintain network documentation: keep network topology and device information documented and updated regularly.
- Perform security testing with authorisation: reconnaissance and scanning should only be performed against systems and networks you own or have written permission to test.

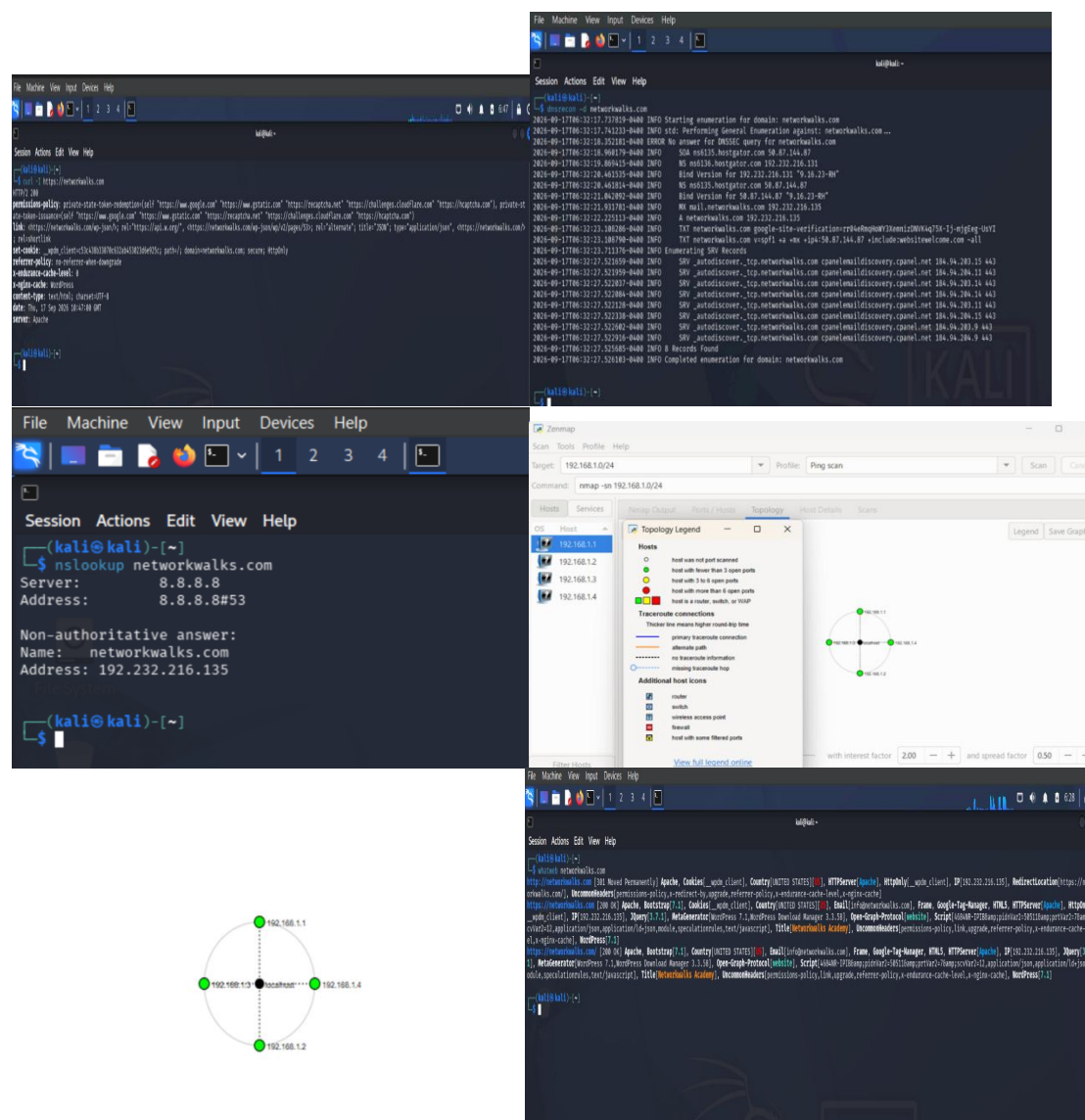
7. Conclusion

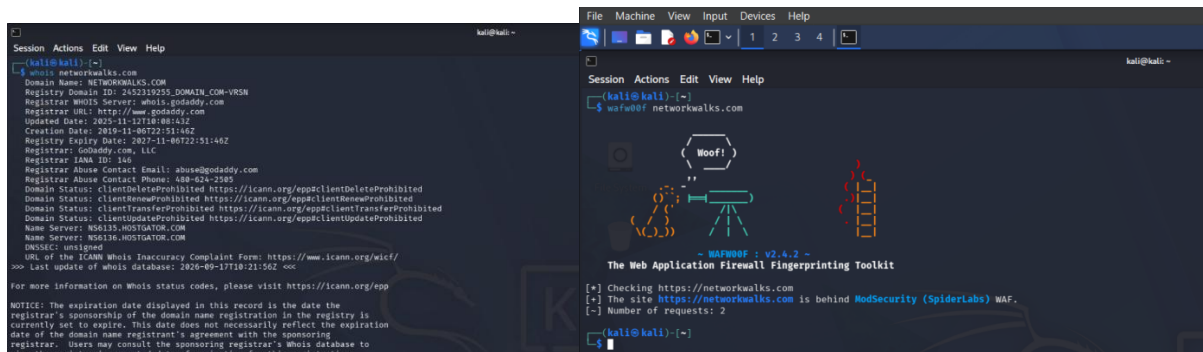
During Week 2 of my Cybersecurity internship at Networkwalks, I completed two practical modules covering footprinting, and network scanning.

In the footprinting module, I used six Kali Linux tools to collect information about the networkwalks.com domain. I learned how WHOIS reveals domain registration details, WhatWeb identifies web technologies, Nslookup resolves domain names, curl inspects HTTP headers, Wafw00f identifies a WAF, and DNSRecon uncovers additional DNS infrastructure information.

In the network scanning module, I used Zenmap to identify my own local network configuration, discover four active hosts, collect their IP and MAC address

Overall, these exercises showed me that information gathering is a critical part of cybersecurity work. Even before attempting to interact with or test a system, a security professional can learn a significant amount about an environment by carefully analysing publicly available information, search-engine indexes, and network responses. I also learned that findings should always be documented clearly: what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk. Finally, I learned that reconnaissance, dork-based searching and scanning must always be performed within an authorised scope; these activities were completed only against my own systems and the Networkwalks domain, for which written permission had already been secured as part of this educational internship program.





-End-

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 **Project Information**

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub